FL Lab2

Friday, 29 September 2023 10:57 PM

Title: Turbulent flow in a pipe

Goal: Determine the friction factor for turbulent flow over a range of flow rates

Theory: Reynold's Number

$$Re = \frac{\rho \langle v \rangle D}{\mu}$$

(ρ: fluid; $\langle v \rangle$: average velocity; D: pipe diameter (ID); μ: viscosity)

$Re > 4000$ (turbulent flow)

(Darcy) Friction Factor (f_D): Twists, turns, pipe flow → frictional losses → can become important in many industrial applications so important to study!

↳ Empirically determined correlations exist

Blasius

$$f_D = 0.316\, Re^{-0.25}$$

Colebrook (surface roughness)

$$\frac{1}{f_D^{1/2}} = -2 \log\left(\frac{\varepsilon/D}{3.7} + \frac{2.51}{Re\, f_D^{1/2}}\right)$$

f_D is non-dimensional!

P.S. f → fanning friction factor $f = \frac{1}{4} f_D$ } Same concept but developed by different scientist so off by a factor!

Darcy-Weisbach Equation

Relates head loss due to friction:

$$\frac{\Delta p}{L} = \frac{f_D \cdot \rho \langle v \rangle^2}{2D} = f_D \left(\rho \frac{\langle v \rangle D}{\mu}\right) \frac{\langle v \rangle}{2D^2} \mu = \frac{f_D \langle v \rangle \mu}{2D^2} (Re)$$

Experiment: + Flow water of specific Q → Record variables in order to calculate Re → Calculate f & compare Darcy vs. (Colebrook & Blasius)

Re: D, $\langle v \rangle$, ρ, μ

Fluid properties: water ✓
Pipe specs: given ✓

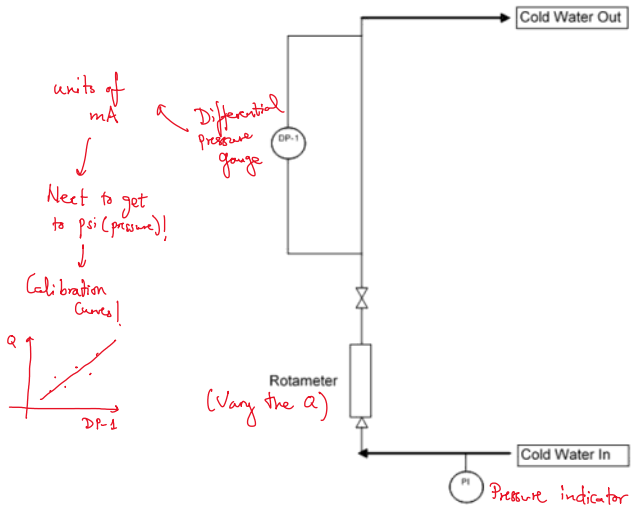
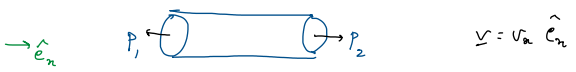


Figure 1: Process Flow Diagram for Turbulent Flow in a Pipe

Pre-Lab (due w/ the lab report due date)

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Hints: ① → RTT on fluid flow in pipe to evaluate $\sum_{cv} f$
→ $\sum_{cv} f$ balance (pressures + shear stress)



$$\sum_{\text{on fluid}} F = \frac{dP}{dt} + \iint_{CS} \rho(\hat{n} \cdot \underline{v})\, \underline{v}\, dS$$

↓ @ SS & CS as opening & exit

$$\sum_{\text{on fluid}} F = \text{Term 1} + \text{Term 2} = 0$$

If $\Sigma F = 0$; the pressures must be balanced w/ shear stress to give net force = 0

$P_1 A - P_2 A - \tau$ (Surface area)

↓

rearrange to give τ & Δp relation

② Similar analysis



redo ① analysis ; now account for gravity as force in force balance

③ Substitution of τ from ①

④ Substitution & analysis